



Dementia with long-term meds

Anticholinergic medicines taken for a long-term period increased the risk of dementia by 50%.
<https://digit.in/july19-58>



Cotton candy explosion

A teacher conducted a dangerous but fun chemistry demonstration with exploding cotton candy. <https://digit.in/july19-59>

cause of Jupiter's massive size, what's now the asteroid belt would have been thrown into turmoil, and this would have sent asteroids flying in towards the sun. A lot of those would have hit earth, and brought water to it. The water on earth is identical to the water found in the asteroid belt – the ratio of deuterium to hydrogen in the water is identical. More testing is needed, and we'd need to properly explore the asteroid belt to confirm this. For now, it's still a mystery waiting to be solved.

WHY IS THERE MATTER?

As per baryon asymmetry, we have a problem explaining why we see so much more matter than antimatter around us. After the Big Bang, there must have been an imbalance in the amount of matter and antimatter, which would have caused what's known as baryogenesis – the birth of matter. Earliest theories suggested that there would have been a very minuscule amount of asymmetry, about 0.00000001% more matter than antimatter. However, experiments done in particle accelerators suggest that the asymmetry might be as high as 1% and that would certainly explain all this matter we have around us. However, why, in accordance with our mathematics, is more matter produced than antimatter is just a mystery that's waiting to be solved.

CONSCIOUSNESS AND FREE WILL

This is a two part mystery, and it's a doozy! The first one is easier to understand for most of us. We are a self-aware species, and we're conscious. But what is consciousness, and where does it reside? What's more important is that we don't fully understand consciousness and where it comes from.

While most biologists will tell you that the consciousness certainly has something to do with the brain, if not everything, we're having a hard time explaining how it is that certain living things that don't have brains are able to accomplish. For example, when you playback the sound of a caterpillar chomping on leaves, certain plants

are able to 'hear' that chomping, and react by letting off chemicals to ward off the caterpillars. How on earth can a plant do this without a brain? Are plants conscious? 'Intelligent Plants' certainly seem to have some low level of consciousness. Controversial ideas for sure, but there is a new field called plant neurobiology that is researching this field. It begs the question, if plants are conscious, and have no brain, then does consciousness really reside in the brain? What is consciousness, and how does it work? Perhaps there is something to the new field of study called quantum consciousness, which aims to use the craziness of the quantum world to describe our consciousness. It's a well researched field, because understanding what consciousness is would probably lead to one being created artificially – and we all know how big an improvement it would be to have proper AI.

A subset of consciousness is the idea of free will. This subset interjects with the chaos, however, it's not just dependent on chaos. Studies have shown that the brain makes up its mind when given a simple choice (yes or no answers) before the person knows the answer themselves. In the study, the researchers were able to predict the choice the volunteer would make up to 11 seconds before they were even asked to choose! Was the 'consciousness' of the participant even involved in the decision-making process? And if it's not a conscious decision, is it really 'free will'. From what we know so far, we know that the brain is definitely involved in the decision making process, but whether or not it needs you (the conscious you) to be involved in the decision is debatable. This has large repercussions, because if there is no such thing as free will, then a murderer could be nothing other than a murderer given his or her life experiences. It would bring into question the idea of the death penalty, and have a lot of us question our very existence. It might even point to us being part of a very grand scale computer simulation, with merely the illusion of free will.

WHAT'S OUTSIDE THE UNIVERSE

Is there an outside to the universe?

We don't know. There are parts of our universe that are forever hidden from us, and if the universe keeps expanding the way it is, even more of the universe will be forever lost to us. The observable universe is about 93 billion light years in diameter, with us at the center. Most of the galaxies inside this universe are at this moment in time accelerating away from us faster than the speed of light because of the spacetime between us expanding that fast. In fact, a much smaller sphere, called the Hubble Sphere is the demarcation of what areas of spacetime aren't moving away from us at faster than light speed. Of course, because the hubble sphere itself is expanding, light emitted from a galaxy that's just outside the Hubble Sphere will eventually reach us billions of years from now. Even today, we're looking at light that left some galaxies billions of years ago, and when they left that galaxy, they were already outside of the hubble sphere of the space we occupy right now, and yet that light reached us because of the expansion of the Hubble Sphere itself because of the expansion of spacetime.

Now, if our universe is expanding, what is it expanding into? For us, that means evoking higher dimensions, and hypothesising about an existence of a multiverse, in which our little universe is merely one of many (perhaps even infinite) bubble universes, all popping in and out of existence on a timescale that's trillions upon trillions of years long. If that is what the truth is, there might be no way for us to ever find out, because our entire species would do really well to last even a million years, let alone a trillion, in order to be there when the universe pops like an over inflated balloon. That raises the question of what the fate of the universe is, which is a mystery as well.

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